

Electromagnetic Radiation

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Chapter Overview

This chapter examines electromagnetic radiation—the generation and propagation of waves from time-varying sources. We begin by introducing the scalar and vector potentials used to compute radiated fields, followed by the application of Green functions to solve the inhomogeneous Helmholtz equations governing these potentials. The general free-space solution and the far-field approximation are then derived, together with the Stratton–Chu formulation. These tools are applied to the Hertzian dipole to analyze its radiation properties and field patterns, after which image theory is introduced. The chapter concludes with the analysis of radiation from an aperture.

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Vector and Scalar Potentials

In the absence of magnetic sources, Gauss' law for magnetism states that $\vec{\nabla} \cdot \vec{B} = 0$. A key vector identity states that the divergence of the curl of any vector field is always zero. Conversely, on the topologically simple free-space domain used here (or locally on a contractible subregion), a smooth divergence-free field can be expressed as a curl. Thus the magnetic flux density $\vec{B}$ may be written in terms of a magnetic vector potential $\vec{A}$:

$$\vec{B} = \vec{\nabla} \times \vec{A}. \quad (3.1.1)$$

Substituting this into Faraday's law (1.4.5) with no magnetic sources ($\vec{M} = 0$), we obtain:

$$\vec{\nabla} \times (\vec{E} + i\omega\vec{A}) = 0. \quad (3.1.2)$$

In a simply connected region, a curl-free vector is the gradient of a scalar function. Choosing that gradient to be $-\vec{\nabla}\varphi$ defines

$$\vec{E} = -\vec{\nabla}\varphi - i\omega\vec{A}. \quad (3.1.3)$$

Vector and Scalar Potentials (cont.)

In a simple medium with no magnetic sources, substituting (3.1.1) and (3.1.3) into Ampère's law (1.4.6) gives

$$\begin{aligned}\frac{1}{\mu} \vec{\nabla} \times \vec{\nabla} \times \vec{A} &= \vec{\nabla} \times \vec{H} = i\omega \vec{D} + \vec{J} \\ &= i\omega\epsilon (-\vec{\nabla}\varphi - i\omega\vec{A}) + \vec{J} \\ &= -i\omega\epsilon \vec{\nabla}\varphi + \omega^2\epsilon\vec{A} + \vec{J}.\end{aligned}$$

Multiplying by μ gives the wave-like equation for $\vec{A}$:

$$\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \mu\vec{J} + \omega^2\mu\epsilon\vec{A} - i\omega\mu\epsilon\vec{\nabla}\varphi. \quad (3.1.4)$$

Substituting (3.1.3) into Gauss' law (1.4.7), with constant scalar permittivity, gives every intermediate term:

$$\begin{aligned}\rho &= \vec{\nabla} \cdot \vec{D} = \epsilon \vec{\nabla} \cdot \vec{E} \\ &= \epsilon \vec{\nabla} \cdot (-\vec{\nabla}\varphi - i\omega\vec{A}) \\ &= -\epsilon \nabla^2\varphi - i\omega\epsilon \vec{\nabla} \cdot \vec{A}.\end{aligned}$$

Vector and Scalar Potentials (cont.)

Dividing by $-\epsilon$ yields the scalar-potential equation

$$\nabla^2 \varphi + i\omega \vec{\nabla} \cdot \vec{A} = -\frac{\rho}{\epsilon}. \quad (3.1.5)$$

Using the vector identity $\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$, and rearranging terms, equations (3.1.4) and (3.1.5) can be rewritten as a pair of coupled wave equations:

$$\nabla^2 \vec{A} + k^2 \vec{A} = -\mu \vec{J} + \vec{\nabla} (\vec{\nabla} \cdot \vec{A} + i\omega \mu \epsilon \varphi), \quad (3.1.6)$$

$$\nabla^2 \varphi + k^2 \varphi = -\rho/\epsilon - i\omega (\vec{\nabla} \cdot \vec{A} + i\omega \mu \epsilon \varphi), \quad (3.1.7)$$

where $k^2 = \omega^2 \mu \epsilon$. The potentials are not unique: for any sufficiently smooth scalar function χ ,

$$\vec{A} \longrightarrow \vec{A} + \vec{\nabla} \chi, \quad \varphi \longrightarrow \varphi - i\omega \chi$$

leaves both $\vec{B} = \vec{\nabla} \times \vec{A}$ and $\vec{E} = -\vec{\nabla} \varphi - i\omega \vec{A}$ unchanged, since

$$\begin{aligned} \vec{\nabla} \times (\vec{A} + \vec{\nabla} \chi) &= \vec{\nabla} \times \vec{A}, \\ -\vec{\nabla}(\varphi - i\omega \chi) - i\omega(\vec{A} + \vec{\nabla} \chi) &= -\vec{\nabla} \varphi - i\omega \vec{A}. \end{aligned}$$

Vector and Scalar Potentials (cont.)

This gauge freedom lets us prescribe the divergence of $\vec{A}$. The Lorenz choice is

$$\vec{\nabla} \cdot \vec{A} + i\omega\mu\epsilon\varphi = 0, \quad (3.1.8)$$

which is the Lorenz gauge condition. With this choice, the equations become decoupled:

$$\nabla^2 \vec{A} + k^2 \vec{A} = -\mu \vec{J}, \quad (3.1.9)$$

$$\nabla^2 \varphi + k^2 \varphi = -\frac{\rho}{\epsilon} = \frac{1}{i\omega\epsilon} \vec{\nabla} \cdot \vec{J}. \quad (3.1.10)$$

When electric charge and electric current are absent, $\vec{\nabla} \cdot \vec{D} = 0$ and, on the same topologically simple domain, it is similarly convenient to define an electric vector potential by

$$\vec{D} = -\vec{\nabla} \times \vec{F}.$$

The minus sign is the convention dual to $\vec{B} = +\vec{\nabla} \times \vec{A}$. Ampere's law then gives the intermediate steps

$$\vec{\nabla} \times \vec{H} = i\omega \vec{D} = -i\omega \vec{\nabla} \times \vec{F} \implies \vec{\nabla} \times (\vec{H} + i\omega \vec{F}) = 0,$$

Vector and Scalar Potentials (cont.)

and hence, in a simply connected region,

$$\vec{H} = -\vec{\nabla}\vartheta - i\omega\vec{F}.$$

The corresponding Lorenz condition is $\vec{\nabla} \cdot \vec{F} + i\omega\mu\epsilon\vartheta = 0$.

The Green Function Method

For the free-space radiation problem below, take $\omega > 0$, real positive μ and ϵ , and the outgoing branch $k = \omega\sqrt{\mu\epsilon} > 0$; then $\eta = \sqrt{\mu/\epsilon} > 0$. The same integral construction extends to a passive homogeneous medium after the corresponding non-growing outgoing branch is selected. Both the scalar potential equation (3.1.10) and the Cartesian components of the vector potential equation (3.1.9) are forms of the inhomogeneous scalar Helmholtz equation. A powerful method for solving such equations is the Green function technique. The Green function G is the solution to the Helmholtz equation for a unit point source (a Dirac delta function):

$$\nabla^2 G + k^2 G = -\delta(\vec{r} - \vec{r}'). \quad (3.1.11)$$

The solution to this equation, representing the field at position $\vec{r}$ due to a point source located at $\vec{r}'$, is given by

$$G(\vec{r}; \vec{r}') = \frac{e^{-ikR}}{4\pi R}, \quad (3.1.12)$$

where $R = |\vec{r} - \vec{r}'|$ is the distance between the source and observation points. The negative sign in the exponential term ensures that the Green function represents an outward-radiating spherical wave. Under the $e^{+i\omega t}$ convention adopted here, an outgoing wave varies as $e^{i(\omega t - kR)}$, that is, with spatial factor e^{-ikR} ; hence the minus sign. This

The Green Function Method (cont.)

form satisfies the *Sommerfeld radiation condition* at infinity, which requires that any radiating field behaves as an outgoing wave that decays with distance:

$$\lim_{R \rightarrow \infty} R (\partial_R G + ikG) = 0. \quad (3.1.13)$$

The opposite exponential sign describes an incoming spherical wave and is excluded by the no-incoming-field boundary condition of the present radiation problem (although incoming waves are, of course, physical when an incident field is prescribed).

The delta function is defined such that:

$$\begin{aligned} \delta(\vec{r} - \vec{r}') &= 0, & \vec{r} &\neq \vec{r}', \\ \int_V \delta(\vec{r} - \vec{r}') dv &= \begin{cases} 1, & \vec{r}' \text{ in the interior of } V, \\ 0, & \vec{r}' \text{ outside } V. \end{cases} \end{aligned} \quad (3.1.14)$$

The Green Function Method (cont.)

The Green function G is the fundamental solution, or impulse response, of the Helmholtz equation. From (3.1.14), equation (3.1.11) can be re-expressed as the following set of conditions on G :

$$\begin{aligned}\nabla^2 G + k^2 G &= 0, \quad \vec{r} \neq \vec{r}', \\ \int_V (\nabla^2 G + k^2 G) dV &= -1, \quad \vec{r}' \text{ in } V.\end{aligned}\tag{3.1.15}$$

We can verify that our solution for G satisfies these conditions. First, for $\vec{r} \neq \vec{r}'$ (i.e., $R \neq 0$):

$$\begin{aligned}\nabla^2 G &= \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial}{\partial R} \frac{e^{-ikR}}{4\pi R} \right) \\ &= -k^2 \frac{e^{-ikR}}{4\pi R} = -k^2 G.\end{aligned}\tag{3.1.16}$$

The Green Function Method (cont.)

Thus, $\nabla^2 G + k^2 G = 0$ for $R \neq 0$. To verify the second condition, we integrate over an infinitesimal spherical volume V_0 centered at $\vec{r}'$ with radius R_0 :

$$\begin{aligned}\int_{V_0} \nabla^2 G dv &= \oint_{S_0} \vec{\nabla} G \cdot d\vec{s} = \oint_{S_0} (\partial_R G|_{R_0}) \hat{R} \cdot d\vec{s} \\ &= -e^{-ikR_0}(1 + ikR_0)\end{aligned}\tag{3.1.17}$$

and

$$\begin{aligned}\int_{V_0} k^2 G dV &= \int_{V_0} \frac{k^2 e^{-ikR}}{4\pi R} R^2 \sin \theta d\theta d\phi dR \\ &= k^2 \int_0^{R_0} R e^{-ikR} dR \\ &= \frac{k^2}{-ik} \int_0^{R_0} R d(e^{-ikR}) \\ &= ik \left[R_0 e^{-ikR_0} + \frac{1}{ik} (e^{-ikR_0} - 1) \right] \\ &= e^{-ikR_0}(ikR_0 + 1) - 1.\end{aligned}\tag{3.1.18}$$

The Green Function Method (cont.)

Adding these two results, we find that $\int_{V_0} (\nabla^2 G + k^2 G) dv = -1$, which confirms that our expression for the Green function is correct.

The General Radiation Solution

By multiplying the Green function equation (3.1.11) by $\mu \vec{J}(\vec{r}')$ and integrating over a volume containing all sources, we obtain:

$$\begin{aligned} & \int_{V'} \mu \vec{J}(\vec{r}') [(\nabla^2 + k^2) G(\vec{r}; \vec{r}')] dv' \\ &= (\nabla^2 + k^2) \int_{V'} \mu \vec{J}(\vec{r}') G(\vec{r}; \vec{r}') dv' \\ &= - \int_{V'} \mu \vec{J}(\vec{r}') \delta(\vec{r} - \vec{r}') dv' = -\mu \vec{J}(\vec{r}). \end{aligned} \tag{3.1.19}$$

By comparing this result with the wave equation for the vector potential (3.1.9), we can express the solution for $\vec{A}(\vec{r})$ as a convolution of the source distribution with the Green function. The reasoning behind this step is superposition. Using the sifting property (3.1.14), the source itself is a continuous superposition of point sources,

$$\mu \vec{J}(\vec{r}) = \int_{V'} \mu \vec{J}(\vec{r}') \delta(\vec{r} - \vec{r}') dv'.$$

The General Radiation Solution (cont.)

The infinitesimal point source at $\vec{r}'$ has vector strength

$$d\vec{S}(\vec{r}') = \mu \vec{J}(\vec{r}') dv'.$$

By (3.1.11), its contribution to the response is

$$d\vec{A}(\vec{r}) = \mu \vec{J}(\vec{r}') G(\vec{r}; \vec{r}') dv'.$$

Since $\nabla^2 + k^2$ is linear, the response to the distribution is the sum of these contributions:

$$\begin{aligned} (\nabla^2 + k^2) \int_{V'} \mu \vec{J}(\vec{r}') G(\vec{r}; \vec{r}') dv' \\ = - \int_{V'} \mu \vec{J}(\vec{r}') \delta(\vec{r} - \vec{r}') dv' = -\mu \vec{J}(\vec{r}), \end{aligned}$$

which is exactly the wave equation (3.1.9). Here the intended problem is radiation by localized sources in an otherwise homogeneous, unbounded medium, with no prescribed incident field. Any second solution differs from this integral by a field $\vec{A}_0$ satisfying $\nabla^2 \vec{A}_0 + k^2 \vec{A}_0 = 0$ everywhere, regular at finite points, and the homogeneous outgoing radiation condition. The exterior uniqueness theorem (componentwise, followed by

The General Radiation Solution (cont.)

continuation through the source-free interior) then gives $\vec{A}_0 = 0$. Without the regularity and no-incident-field assumptions, an arbitrary homogeneous background field could be added. Under the stated radiation assumptions, the unique solution is

$$\vec{A}(\vec{r}) = \int_{V'} \mu \vec{J}(\vec{r}') G(\vec{r}; \vec{r}') dv'. \quad (3.1.20)$$

Similarly, the continuity equation for the $e^{+i\omega t}$ convention is

$$\vec{\nabla}' \cdot \vec{J}(\vec{r}') = -i\omega\rho(\vec{r}'), \quad \rho = -\frac{1}{i\omega} \vec{\nabla}' \cdot \vec{J}.$$

Here $\vec{\nabla}$ acts on the observation coordinate $\vec{r}$, whereas $\vec{\nabla}'$ acts on the source coordinate $\vec{r}'$. Therefore (3.1.11) and (3.1.10) give the equivalent scalar-potential forms

$$\varphi(\vec{r}) = \frac{1}{\epsilon} \int_{V'} \rho(\vec{r}') G dv' = -\frac{1}{i\omega\epsilon} \int_{V'} (\vec{\nabla}' \cdot \vec{J}(\vec{r}')) G dv'. \quad (3.1.21)$$

The General Radiation Solution (cont.)

Differentiation with respect to $\vec{r}$ may be moved under the source integral. The two terms of (3.1.3) are then

$$\begin{aligned}-\vec{\nabla}\varphi &= \frac{1}{i\omega\epsilon} \int_{V'} (\vec{\nabla}' \cdot \vec{J}) \vec{\nabla} G dv', \\ -i\omega\vec{A} &= -i\omega\mu \int_{V'} \vec{J} G dv'.\end{aligned}$$

Using

$$\omega\mu = k\eta, \quad \frac{1}{i\omega\epsilon} = -\frac{i}{\omega\epsilon} = -\frac{ik\eta}{k^2},$$

and adding the two contributions gives the electric radiation integral

$$\begin{aligned}\vec{E}(\vec{r}) &= -ik\eta \int_{V'} \left[\vec{J}(\vec{r}') G(\vec{r}; \vec{r}') \right. \\ &\quad \left. + \frac{1}{k^2} (\vec{\nabla}' \cdot \vec{J}(\vec{r}')) \vec{\nabla} G(\vec{r}; \vec{r}') \right] dv'.\end{aligned}\tag{3.1.22}$$

The General Radiation Solution (cont.)

For the magnetic field, the product rule and $\vec{\nabla} \times (\vec{J}G) = \vec{\nabla}G \times \vec{J} = -\vec{J} \times \vec{\nabla}G$ give every step:

$$\begin{aligned}\vec{H}(\vec{r}) &= \frac{1}{\mu} \vec{\nabla} \times \vec{A} = \int_{V'} \vec{\nabla} \times (\vec{J}(\vec{r}')G) dV' \\ &= - \int_{V'} \vec{J}(\vec{r}') \times \vec{\nabla}G(\vec{r}; \vec{r}') dV'. \end{aligned} \quad (3.1.23)$$

It is convenient to define the following integral operators:

$$\mathcal{L}\{\vec{X}\} = -ik \int_{V'} \left[\vec{X}G + \frac{1}{k^2}(\vec{\nabla}' \cdot \vec{X})\vec{\nabla}G \right] dV', \quad (3.1.24)$$

$$\mathcal{R}\{\vec{X}\} = - \int_{V'} \vec{X} \times \vec{\nabla}G dV'. \quad (3.1.25)$$

This expanded form keeps the operator placement visible: $\vec{\nabla}'$ differentiates the source distribution, whereas $\vec{\nabla}$ differentiates G . A notation such as $\vec{\nabla}[(\vec{\nabla}' \cdot \vec{X})G]$ is equivalent only when it is understood that the observation gradient passes through the source-only factor and acts on G :

$$\vec{\nabla} [(\vec{\nabla}' \cdot \vec{X})G] = (\vec{\nabla}' \cdot \vec{X})\vec{\nabla}G.$$

The General Radiation Solution (cont.)

To make the duality signs explicit, write Maxwell's curl equations with both source types as

$$\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H} - \vec{M}, \quad \vec{\nabla} \times \vec{H} = i\omega\epsilon\vec{E} + \vec{J}.$$

Let $(\vec{E}_X, \vec{H}_X)$ be the fields generated by an electric current $\vec{X}$ with no magnetic current. Define

$$\vec{E}_d = -\eta\vec{H}_X, \quad \vec{H}_d = \frac{1}{\eta}\vec{E}_X, \quad \vec{M}_d = \eta\vec{X}, \quad \vec{J}_d = \vec{0}.$$

Using $\eta\epsilon = \mu/\eta = \sqrt{\mu\epsilon}$, direct substitution gives

$$\begin{aligned}\vec{\nabla} \times \vec{E}_d &= -\eta (i\omega\epsilon\vec{E}_X + \vec{X}) = -i\omega\mu\vec{H}_d - \vec{M}_d, \\ \vec{\nabla} \times \vec{H}_d &= \frac{1}{\eta} (-i\omega\mu\vec{H}_X) = i\omega\epsilon\vec{E}_d.\end{aligned}$$

For a prescribed magnetic current $\vec{M}$, choose $\vec{X} = \vec{M}/\eta$. Inserting the electric-source fields $\vec{E}_X = \eta\mathcal{L}\{\vec{X}\}$ and $\vec{H}_X = \mathcal{R}\{\vec{X}\}$ then yields the magnetic-source formulas below without an implicit sign choice. Using these operators, the electric and magnetic fields

The General Radiation Solution (cont.)

generated by electric and magnetic sources can be expressed in a compact and symmetric form. For electric sources:

$$\vec{E} = \eta \mathcal{L}\{\vec{J}\}, \quad (3.1.26)$$

$$\vec{H} = \mathcal{K}\{\vec{J}\}, \quad (3.1.27)$$

and by applying the duality transform, for magnetic sources:

$$\vec{E} = -\mathcal{K}\{\vec{M}\}, \quad (3.1.28)$$

$$\vec{H} = \frac{1}{\eta} \mathcal{L}\{\vec{M}\}. \quad (3.1.29)$$

By the principle of superposition, the total fields from both electric and magnetic sources are:

$$\vec{E} = \eta \mathcal{L}\{\vec{J}\} - \mathcal{K}\{\vec{M}\}, \quad (3.1.30)$$

$$\vec{H} = \mathcal{K}\{\vec{J}\} + \frac{1}{\eta} \mathcal{L}\{\vec{M}\}. \quad (3.1.31)$$

The General Radiation Solution (cont.)

An alternative expression follows directly from the Lorenz condition:

$$\varphi = -\frac{1}{i\omega\mu\epsilon}\vec{\nabla}\cdot\vec{A},$$

and therefore

$$\begin{aligned}\vec{E} &= -i\omega\vec{A} + \frac{1}{i\omega\mu\epsilon}\vec{\nabla}(\vec{\nabla}\cdot\vec{A}) \\ &= -i\omega\left(1 + \frac{1}{k^2}\vec{\nabla}\vec{\nabla}\cdot\right)\vec{A}.\end{aligned}$$

Substitution of (3.1.20) and use of $\omega\mu = k\eta$ now gives

$$\vec{E}(\vec{r}) = -ik\eta \int_{V'} \left(1 + \frac{1}{k^2}\vec{\nabla}\vec{\nabla}\cdot\right) [\vec{J}(\vec{r}')G(\vec{r};\vec{r}')] dv'. \quad (3.1.32)$$

It is important to understand the distinction between the two expressions for the electric field, (3.1.22) and (3.1.32). In (3.1.22), $\vec{\nabla}'$ differentiates the source current and $\vec{\nabla}$ differentiates only the Green function. This results in a weaker singularity in the integrand. In contrast, in (3.1.32), both $\vec{\nabla}$ operators act on the field point $\vec{r}$, and therefore on the Green function, leading to a higher-order singularity. This form is typically more suitable for far-field calculations, where simplifying approximations can be made.

The Stratton-Chu Formulation

Based on the surface equivalence principle (Section 1.5.4), let a closed surface S_0 lie in a homogeneous, source-free neighborhood and enclose all actual sources. Choose $\hat{n}$ to point from the extinguished interior into the exterior region of interest. Setting the equivalent interior fields to zero replaces the original sources by

$$\begin{cases} \vec{J}_s = \hat{n} \times \vec{H}, \\ \vec{M}_s = -\hat{n} \times \vec{E}, \end{cases} \quad (3.1.33)$$

where the fields on the right are the original exterior limiting fields. For an observation point strictly outside S_0 , the general radiation solution (3.1.30) gives

$$\begin{aligned} \vec{E} &= \eta \mathcal{L}\{\vec{J}_s\} - \mathcal{R}\{\vec{M}_s\} \\ &= -ik\eta \oint_{S_0} \left[\vec{J}_s G + \frac{1}{k^2} (\vec{\nabla}'_s \cdot \vec{J}_s) \vec{\nabla} G \right] ds' \\ &\quad + \oint_{S_0} \vec{M}_s \times \vec{\nabla} G ds'. \end{aligned} \quad (3.1.34)$$

The Stratton-Chu Formulation (cont.)

Here $\vec{\nabla}'_s \cdot$ is the surface divergence. Because there is no normal volume-current flux on either side of the equivalent sheet, surface charge conservation and the normal- $\vec{D}$ matching condition (1.2.8) give, step by step,

$$\vec{\nabla}'_s \cdot \vec{J}_s + i\omega\rho_s = 0, \quad \rho_s = \hat{n} \cdot (\vec{D}_{\text{out}} - \vec{D}_{\text{in}}) = \epsilon \hat{n} \cdot \vec{E}, \quad (3.1.35)$$

or

$$\vec{\nabla}'_s \cdot \vec{J}_s = -i\omega\epsilon(\hat{n} \cdot \vec{E}).$$

Substituting (3.1.33) and (3.1.35) into (3.1.34) and using the property $\vec{\nabla}' G = -\vec{\nabla} G$, we arrive at the expression for the electric field:

$$\begin{aligned} -\frac{ik\eta}{k^2} [-i\omega\epsilon(\hat{n} \cdot \vec{E})] \vec{\nabla} G &= -(\hat{n} \cdot \vec{E}) \vec{\nabla} G = (\hat{n} \cdot \vec{E}) \vec{\nabla}' G, \\ \vec{M}_s \times \vec{\nabla} G &= (-\hat{n} \times \vec{E}) \times (-\vec{\nabla}' G) \\ &= (\hat{n} \times \vec{E}) \times \vec{\nabla}' G, \end{aligned}$$

The Stratton-Chu Formulation (cont.)

where $k\eta = \omega\mu$ and $k^2 = \omega^2\mu\epsilon$ make the scalar coefficient in the first line equal to -1 . Therefore

$$\begin{aligned}\vec{E} = \oint_{S_0} & \left[-ik\eta (\hat{n} \times \vec{H}) G + (\hat{n} \cdot \vec{E}) \vec{\nabla}' G \right. \\ & \left. + (\hat{n} \times \vec{E}) \times \vec{\nabla}' G \right] ds'.\end{aligned}\tag{3.1.36}$$

By applying the duality transform, we can obtain the corresponding expression for the magnetic field:

$$\begin{aligned}\vec{H} = \oint_{S_0} & \left[i\frac{k}{\eta} (\hat{n} \times \vec{E}) G + (\hat{n} \cdot \vec{H}) \vec{\nabla}' G \right. \\ & \left. + (\hat{n} \times \vec{H}) \times \vec{\nabla}' G \right] ds'.\end{aligned}\tag{3.1.37}$$

This pair of equations constitutes the *Stratton–Chu formulation*, which provides a rigorous and general framework for solving radiation problems. Notably, the Stratton–Chu formulation explicitly involves the normal components of the electric and magnetic fields.

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The Far-Field Approximation

In the far-field region, the observation distance r is much larger than the source size $a = \max |\vec{r}'|$, and the observation point is many wavelengths away, $kr \gg 1$. For an extended source, the phase approximation below also requires $ka^2/r \ll 1$, the Fraunhofer condition. The condition $r \gg a$ controls the amplitude expansion, while $kr \gg 1$ controls the neglect of higher-order $1/r$ field terms. The denominator of the Green function (3.1.12) can be approximated as $4\pi r$, and the phase term in the numerator can be expanded as

$$\begin{aligned} R = |\vec{r} - \vec{r}'| &= \sqrt{r^2 - 2r(\hat{r} \cdot \vec{r}') + r'^2} \\ &= r \left(1 - 2\frac{\hat{r} \cdot \vec{r}'}{r} + \frac{r'^2}{r^2} \right)^{1/2}. \end{aligned}$$

Using the generalized binomial theorem with small argument x ,

$$(1 + x)^{1/2} \approx 1 + \frac{1}{2}x,$$

and retaining only the first-order term, we obtain

$$R \approx r - \hat{r} \cdot \vec{r}'. \quad (3.2.1)$$

The Far-Field Approximation (cont.)

The approximation (3.2.1) is first order in the source-size ratio a/r . If the next term is needed, the binomial expansion must be carried one order further. With

$$x = -2\frac{\hat{r} \cdot \vec{r}'}{r} + \frac{r'^2}{r^2},$$

substitution into $(1+x)^{1/2} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2$ gives, after multiplying by r ,

$$R \approx r - \hat{r} \cdot \vec{r}' + \frac{r'^2 - (\hat{r} \cdot \vec{r}')^2}{2r} + \dots,$$

so the term dropped in (3.2.1) is $O(a^2/r)$. The amplitude factor expands as

$$\frac{1}{R} = \frac{1}{r} \left[1 + \frac{\hat{r} \cdot \vec{r}'}{r} + O\left(\frac{a^2}{r^2}\right) \right].$$

Its relative correction is $O(a/r)$, so $1/R$ may be replaced by $1/r$ when $r \gg a$. The phase term e^{-ikR} is more delicate, because the error in R is multiplied by k . The omitted

The Far-Field Approximation (cont.)

quadratic term contributes a phase error of order ka^2/r , so the linear phase approximation is valid when $ka^2/r \ll 1$. Under this condition,

$$e^{-ikR} \approx e^{-ikr(1-\hat{\mathbf{r}} \cdot \vec{\mathbf{r}}'/r)} \approx e^{-ikr} e^{ik\hat{\mathbf{r}} \cdot \vec{\mathbf{r}}'}. \quad (3.2.2)$$

This approximation preserves the leading angular dependence of the radiation pattern. Thus, in the far field, the Green function simplifies to

$$G(\vec{\mathbf{r}}; \vec{\mathbf{r}}') \approx \frac{e^{-ikr}}{4\pi r} e^{ik\hat{\mathbf{r}} \cdot \vec{\mathbf{r}}'} = G_r(r) G_a(\theta, \phi), \quad (3.2.3)$$

where $G_r(r) = e^{-ikr}/(4\pi r)$ represents the radial dependence, and $G_a(\theta, \phi) = e^{ik\hat{\mathbf{r}} \cdot \vec{\mathbf{r}}'}$ represents the angular dependence.

Here $k = 2\pi/\lambda$. To apply (3.2.3) to the radiation integral, differentiate both factors without suppressing their orders. Since

$$\vec{\nabla} \hat{\mathbf{r}} = \frac{\vec{\mathbf{l}} - \hat{\mathbf{r}} \hat{\mathbf{r}}}{r}, \quad \vec{\nabla}(\hat{\mathbf{r}} \cdot \vec{\mathbf{r}}') = \frac{\vec{\mathbf{l}} - \hat{\mathbf{r}} \hat{\mathbf{r}}}{r} \cdot \vec{\mathbf{r}}',$$

The Far-Field Approximation (cont.)

we have

$$\begin{aligned}\vec{\nabla} G_r &= \left(-ik - \frac{1}{r}\right) G_r \hat{r}, \\ \vec{\nabla} G_a &= \frac{ik}{r} G_a \left(\vec{l} - \hat{r}\hat{r}\right) \cdot \vec{r}'.\end{aligned}$$

The second line has magnitude at most $ka|G_a|/r$. The product rule therefore gives the dimensionally explicit result

$$\begin{aligned}\vec{\nabla} G &= G \left[-ik\hat{r} - \frac{\hat{r}}{r} + \frac{ik}{r} \left(\vec{l} - \hat{r}\hat{r}\right) \cdot \vec{r}'\right] \\ &= -ikG\hat{r} + G O\left(\frac{1+ka}{r}\right) \approx -ikG\hat{r}.\end{aligned}\tag{3.2.4}$$

Because $G = O(1/r)$, the absolute remainder is $O((1+ka)/r^2)$, whereas the retained term is $O(k/r)$. Thus $kr \gg 1$ and $r \gg a$ make the derivative approximation uniform over the bounded source.

Next, define the displayed gradient remainder by

$$\vec{\nabla} G = -ikG\hat{r} + \vec{\Delta}, \quad \vec{\Delta} = G O\left(\frac{1+ka}{r}\right).$$

The Far-Field Approximation (cont.)

Because $\vec{J}(\vec{r}')$ is independent of the observation coordinate,

$$\begin{aligned}\vec{\nabla} \vec{\nabla} \cdot [\vec{J}G] &= \vec{\nabla}[\vec{J} \cdot \vec{\nabla}G] \\ &= \vec{\nabla}[-ik(\hat{r} \cdot \vec{J})G + \vec{J} \cdot \vec{\Delta}] \\ &= -ik \{ (\hat{r} \cdot \vec{J})\vec{\nabla}G + G\vec{\nabla}(\hat{r} \cdot \vec{J}) \} + \vec{\nabla}(\vec{J} \cdot \vec{\Delta}).\end{aligned}\tag{3.2.5}$$

To proceed with this derivation, we first note the following vector identity involving the unit dyadic $\vec{\vec{I}}$:

$$\begin{aligned}\vec{\nabla} \vec{r} &= (\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z)(\hat{x}x + \hat{y}y + \hat{z}z) \\ &= \hat{x}\hat{x} + \hat{y}\hat{y} + \hat{z}\hat{z} = \vec{\vec{I}},\end{aligned}\tag{3.2.6}$$

where a dyadic is a tensor of rank two, which can be represented as a square matrix:

$$\begin{aligned}\vec{\vec{F}} &= F_{xx}\hat{x}\hat{x} + F_{yx}\hat{y}\hat{x} + F_{zx}\hat{z}\hat{x} \\ &\quad + F_{xy}\hat{x}\hat{y} + F_{yy}\hat{y}\hat{y} + F_{zy}\hat{z}\hat{y} \\ &\quad + F_{xz}\hat{x}\hat{z} + F_{yz}\hat{y}\hat{z} + F_{zz}\hat{z}\hat{z}\end{aligned}$$

The Far-Field Approximation (cont.)

$$= \begin{bmatrix} F_{xx} & F_{xy} & F_{xz} \\ F_{yx} & F_{yy} & F_{yz} \\ F_{zx} & F_{zy} & F_{zz} \end{bmatrix}. \quad (3.2.7)$$

The juxtaposition of two vectors, such as $\vec{\vec{F}} = \vec{a}\vec{b}$, forms a dyadic product with components $F_{mn} = a_m b_n$. We have the following rules for dyadic calculations:

- $\vec{c} \cdot (\vec{a}\vec{b}) = (\vec{c} \cdot \vec{a})\vec{b}$
- $(\vec{a}\vec{b}) \cdot \vec{c} = \vec{a}(\vec{b} \cdot \vec{c})$
- $\vec{c} \times (\vec{a}\vec{b}) = (\vec{c} \times \vec{a})\vec{b}$
- $(\vec{a}\vec{b}) \times \vec{c} = \vec{a}(\vec{b} \times \vec{c})$

The Far-Field Approximation (cont.)

From the identity $\vec{\nabla} \vec{r} = \vec{\nabla}(r\hat{r}) = \vec{\nabla}(r)\hat{r} + r\vec{\nabla}(\hat{r}) = \hat{r}\hat{r} + r\vec{\nabla}(\hat{r}) = \vec{\bar{l}}$, we can derive:

$$\vec{\nabla}\hat{r} = \frac{\vec{\bar{l}} - \hat{r}\hat{r}}{r}. \quad (3.2.8)$$

The term $\vec{\nabla}[\hat{r} \cdot \vec{J}(\vec{r}')]]$ in (3.2.5) can be simplified using the vector identity $\vec{\nabla}(\vec{a} \cdot \vec{b}) = \vec{a} \times (\vec{\nabla} \times \vec{b}) + \vec{b} \times (\vec{\nabla} \times \vec{a}) + (\vec{a} \cdot \vec{\nabla})\vec{b} + (\vec{b} \cdot \vec{\nabla})\vec{a}$:

$$\begin{aligned} \vec{\nabla} [\hat{r} \cdot \vec{J}(\vec{r}')]] &= [\vec{J}(\vec{r}') \cdot \vec{\nabla}] \hat{r} = \vec{J}(\vec{r}') \cdot \vec{\nabla} \hat{r} \\ &= \frac{\vec{J} - J_r \hat{r}}{r} = \frac{J_\theta \hat{\theta} + J_\phi \hat{\phi}}{r} = O\left(\frac{1}{r}\right). \end{aligned} \quad (3.2.9)$$

Thus, with $\vec{J}_T = \vec{J} - J_r \hat{r}$, the expression in (3.2.5) becomes

$$\begin{aligned} \vec{\nabla} \vec{\nabla} \cdot [\vec{J}G] &= -ik \left\{ J_r (-ikG\hat{r} + \vec{\Delta}) + G \frac{\vec{J}_T}{r} \right\} + \vec{\nabla}(\vec{J} \cdot \vec{\Delta}) \\ &= -k^2 J_r \hat{r} G \\ &\quad + O\left(\frac{kG(1+ka)|\vec{J}|}{r}\right) + O\left(\frac{G(1+ka)^2|\vec{J}|}{r^2}\right). \end{aligned} \quad (3.2.10)$$

The Far-Field Approximation (cont.)

Relative to the retained $k^2 G[\vec{J}]$ term, the displayed errors are of order $(1 + ka)/(kr)$ and its square; they vanish under the far-field conditions stated above. Applying these approximations, the expression for the electric field in (3.1.32) simplifies to:

$$\begin{aligned}\vec{E} &\approx -ik\eta \int_{V'} G [\vec{J}(\vec{r}') - J_r \hat{r}] dv' \\ &= -ik\eta \frac{e^{-ikr}}{4\pi r} \int_{V'} (J_\theta \hat{\theta} + J_\phi \hat{\phi}) e^{ik\hat{r} \cdot \vec{r}'} dv' \\ &= -ik\eta \frac{e^{-ikr}}{4\pi r} \int_{V'} (\hat{\theta} \hat{\theta} + \hat{\phi} \hat{\phi}) \cdot \vec{J}(\vec{r}') e^{ik\hat{r} \cdot \vec{r}'} dv' \\ &= ik\eta \frac{e^{-ikr}}{4\pi r} \int_{V'} \hat{r} \times [\hat{r} \times \vec{J}(\vec{r}')] e^{ik\hat{r} \cdot \vec{r}'} dv'.\end{aligned}\tag{3.2.11}$$

From (3.1.23), the magnetic field in the far-field region is given by:

$$\vec{H} \approx -ik \frac{e^{-ikr}}{4\pi r} \int_{V'} [\hat{r} \times \vec{J}(\vec{r}')] e^{ik\hat{r} \cdot \vec{r}'} dv' = \frac{1}{\eta} \hat{r} \times \vec{E}.\tag{3.2.12}$$

The Far-Field Approximation (cont.)

For the general case including both electric and magnetic sources, we can apply the duality principle to obtain the far-field expression for the electric field and the magnetic field:

$$\vec{E} \approx -ik \frac{e^{-ikr}}{4\pi r} \int_{V'} [\eta (\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{J}(\vec{r}') - \hat{r} \times \vec{M}(\vec{r}')] e^{ik\hat{r} \cdot \vec{r}'} dv' \quad (3.2.13)$$

$$\vec{H} \approx -ik \frac{e^{-ikr}}{4\pi r} \int_{V'} \left[\frac{1}{\eta} (\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{M}(\vec{r}') + \hat{r} \times \vec{J}(\vec{r}') \right] e^{ik\hat{r} \cdot \vec{r}'} dv'. \quad (3.2.14)$$

If the spatial Fourier transform is defined by

$$\tilde{J}(\vec{q}) = \int \vec{J}(\vec{r}') e^{i\vec{q} \cdot \vec{r}'} dv',$$

then the electric-current far field samples its transverse projection on the sphere $\vec{q} = k\hat{r}$. With the opposite sign convention this same integral is called the inverse transform; the physically invariant statement is the sampling and transverse projection, not the

The Far-Field Approximation (cont.)

transform name. The three equalities in the chain leading to (3.2.13) rest on the resolution of the source current into its radial and transverse parts. Writing $\vec{J}(\vec{r}') = J_r \hat{r} + J_\theta \hat{\theta} + J_\phi \hat{\phi}$ with $J_r = \hat{r} \cdot \vec{J}$, the radial component is removed by subtraction,

$$\vec{J}(\vec{r}') - J_r \hat{r} = J_\theta \hat{\theta} + J_\phi \hat{\phi},$$

which is the purely transverse current already encountered in the simplification of the radiation integral. Recalling that the unit dyadic resolves in the spherical basis as

$\bar{\bar{I}} = \hat{r}\hat{r} + \hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}$, this transverse current is the action of the projection dyadic $\bar{\bar{I}} - \hat{r}\hat{r} = \hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}$ on $\vec{J}$; using the dyadic rule $(\vec{a}\vec{b}) \cdot \vec{c} = \vec{a}(\vec{b} \cdot \vec{c})$,

$$\begin{aligned}(\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{J} &= \hat{\theta}(\hat{\theta} \cdot \vec{J}) + \hat{\phi}(\hat{\phi} \cdot \vec{J}) \\ &= J_\theta \hat{\theta} + J_\phi \hat{\phi}.\end{aligned}$$

Finally, the same transverse current may be written as a double cross product. Applying the identity $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$ with $\vec{a} = \vec{b} = \hat{r}$ and $\hat{r} \cdot \hat{r} = 1$,

$$\begin{aligned}\hat{r} \times (\hat{r} \times \vec{J}) &= \hat{r}(\hat{r} \cdot \vec{J}) - \vec{J} = J_r \hat{r} - \vec{J} \\ &= -(\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{J},\end{aligned}$$

The Far-Field Approximation (cont.)

so the transverse projection equals $-\hat{r} \times (\hat{r} \times \vec{J})$, and the sign of the prefactor reverses from $-ik\eta$ to $+ik\eta$ in the last line.

The mixed-source form (3.2.13) then follows by duality. For magnetic sources $\vec{E} = -\Re\{\vec{M}\}$, and with $\vec{\nabla} G \approx -ikG\hat{r}$ in the far field,

$$\begin{aligned}\Re\{\vec{M}\} &= - \int_{V'} \vec{M} \times \vec{\nabla} G \, dv' \approx ik \int_{V'} (\vec{M} \times \hat{r}) G \, dv' \\ &= -ik \frac{e^{-ikr}}{4\pi r} \int_{V'} (\hat{r} \times \vec{M}) e^{ik\hat{r} \cdot \vec{r}'} \, dv' .\end{aligned}$$

Hence $\vec{E} = -\Re\{\vec{M}\}$ contributes a term $+ik(e^{-ikr}/4\pi r)\hat{r} \times \vec{M}$, which appears as $-\hat{r} \times \vec{M}$ inside the bracket of (3.2.13) once the common factor $-ik$ is pulled out; superposing this with the electric contribution, and repeating the argument with the roles of $\vec{J}$ and $\vec{M}$ exchanged, yields (3.2.14).

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Radiation from a Hertzian Dipole

The Hertzian dipole is the ideal point-current model of an electrically short, uniform-current element. Let its effective length be denoted by dl , let q be the endpoint-charge phasor, and let positive current flow toward the endpoint carrying $+q$. The time-domain relation $I(t) = dq(t)/dt$ becomes, under the $e^{+i\omega t}$ convention,

$$I = i\omega q, \quad \vec{p} = q dl \hat{z}, \quad I dl \hat{z} = i\omega \vec{p}.$$

The point-source distribution at the origin is

$$\vec{J}(\vec{r}') = I dl \hat{z} \delta(\vec{r}'), \quad \int \vec{J}(\vec{r}') dv' = I dl \hat{z}.$$

Equivalently, before the infinitesimal limit a filament on the z -axis has $\vec{J} = I \hat{z} \delta(x') \delta(y')$ over the element, so $\vec{J} dv' = I \hat{z} dz'$; integration over its length produces $I dl \hat{z}$. The associated endpoint charges are required by the continuity equation, so the ideal current element is not a current source in isolation. Substitution into (3.1.32) gives

$$\vec{E}(\vec{r}) = -ik\eta I dl \left(1 + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \cdot\right) \hat{z} G. \quad (3.3.1)$$

Radiation from a Hertzian Dipole (cont.)

and (3.1.23) gives

$$\vec{H}(\vec{r}) = -Idl \hat{z} \times \vec{\nabla} G. \quad (3.3.2)$$

To evaluate the derivatives in spherical coordinates, first differentiate $G = e^{-ikr}/(4\pi r)$ explicitly:

$$\begin{aligned} \partial_r G &= \frac{1}{4\pi} \left[(-ik) e^{-ikr} \frac{1}{r} + e^{-ikr} \left(-\frac{1}{r^2} \right) \right] \\ &= -\left(ik + \frac{1}{r} \right) G, \end{aligned} \quad (3.3.3)$$

$$\vec{\nabla} G = \hat{r} \partial_r G = -\left(ik + \frac{1}{r} \right) G \hat{r}.$$

Because $\hat{z}$ is a constant Cartesian vector and $\hat{z} \cdot \hat{r} = \cos \theta$,

$$\psi \equiv \vec{\nabla} \cdot (\hat{z} G) = \hat{z} \cdot \vec{\nabla} G = -\left(ik + \frac{1}{r} \right) G \cos \theta.$$

Write $\psi = g(r) \cos \theta$, where

$$g(r) = -\frac{1}{4\pi} \left(\frac{ik}{r} + \frac{1}{r^2} \right) e^{-ikr}.$$

Radiation from a Hertzian Dipole (cont.)

The radial differentiation is

$$\begin{aligned} g'(r) &= -\frac{e^{-ikr}}{4\pi} \left[-\frac{ik}{r^2} - \frac{2}{r^3} + \left(\frac{ik}{r} + \frac{1}{r^2} \right) (-ik) \right] \\ &= -\frac{e^{-ikr}}{4\pi} \left[\frac{k^2}{r} - \frac{2ik}{r^2} - \frac{2}{r^3} \right] \\ &= G \left(-k^2 + \frac{2ik}{r} + \frac{2}{r^2} \right), \end{aligned}$$

whereas the polar differentiation is

$$\frac{1}{r} \partial_\theta \psi = -\frac{g(r)}{r} \sin \theta = \frac{1}{r} \left(ik + \frac{1}{r} \right) G \sin \theta.$$

Thus no derivative step is implicit:

$$\begin{aligned} \vec{\nabla} [\vec{\nabla} \cdot (\hat{z}G)] &= G \left(-k^2 + \frac{2ik}{r} + \frac{2}{r^2} \right) \cos \theta \hat{r} \\ &\quad + \frac{G}{r} \left(ik + \frac{1}{r} \right) \sin \theta \hat{\theta}. \end{aligned} \tag{3.3.4}$$

Radiation from a Hertzian Dipole (cont.)

The cross product needed for the magnetic field is

$$\begin{aligned}\hat{z} \times \vec{\nabla} G &= (\cos \theta \hat{r} - \sin \theta \hat{\theta}) \times \left[- \left(ik + \frac{1}{r} \right) G \hat{r} \right] \\ &= - \left(ik + \frac{1}{r} \right) G \sin \theta \hat{\phi},\end{aligned}$$

where $\hat{\theta} \times \hat{r} = -\hat{\phi}$.

For the electric field, resolve $\hat{z}G = G(\cos \theta \hat{r} - \sin \theta \hat{\theta})$. The radial part inside the operator is

$$\begin{aligned}\left[\hat{z}G + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \cdot (\hat{z}G) \right]_{\hat{r}} &= G \cos \theta \left[1 - 1 + \frac{2i}{kr} + \frac{2}{k^2 r^2} \right] \\ &= G \cos \theta \left(\frac{2i}{kr} + \frac{2}{k^2 r^2} \right).\end{aligned}$$

Multiplication by $-ik\eta l \, dl$ gives

$$E_r = \frac{2\eta l \, dl}{r} \left(1 + \frac{1}{ikr} \right) G \cos \theta.$$

Radiation from a Hertzian Dipole (cont.)

Similarly, the polar part is

$$\begin{aligned}\left[\hat{z}G + \frac{1}{k^2}\vec{\nabla}\vec{\nabla} \cdot (\hat{z}G)\right]_{\hat{\theta}} &= G \sin \theta \left[-1 + \frac{1}{k^2 r} \left(ik + \frac{1}{r}\right)\right] \\ &= G \sin \theta \left(-1 + \frac{i}{kr} + \frac{1}{k^2 r^2}\right),\end{aligned}$$

and hence

$$E_{\theta} = ik\eta I dl \left(1 + \frac{1}{ikr} - \frac{1}{k^2 r^2}\right) G \sin \theta.$$

Combining the components, the exact fields away from the point source are

$$\begin{aligned}\vec{E} &= \frac{2\eta I dl}{r} \left(1 + \frac{1}{ikr}\right) \cos \theta G \hat{r} \\ &\quad + ik\eta I dl \left(1 + \frac{1}{ikr} - \frac{1}{k^2 r^2}\right) \sin \theta G \hat{\theta},\end{aligned}\tag{3.3.5}$$

$$\begin{aligned}\vec{H} &= -I dl \hat{z} \times \vec{\nabla} G \\ &= I dl \left(ik + \frac{1}{r}\right) G \sin \theta \hat{\phi} \\ &= ikI dl \left(1 + \frac{1}{ikr}\right) G \sin \theta \hat{\phi}.\end{aligned}\tag{3.3.6}$$

Radiation from a Hertzian Dipole (cont.)

These expressions include terms that scale as $1/r$, $1/r^2$, and $1/r^3$, corresponding to radiative, inductive, and electrostatic components, respectively. This decomposition enables classification of the field into near-field ($kr \ll 1$) and far-field ($kr \gg 1$) regions. In the near field, set $G \approx 1/(4\pi r)$ and retain the highest inverse power in each component:

$$\begin{aligned}E_r &\approx \frac{2\eta l dl}{r} \frac{1}{ikr} \frac{\cos \theta}{4\pi r} = -\frac{i\eta l dl}{4\pi kr^3} 2 \cos \theta, \\E_\theta &\approx ik\eta l dl \left(-\frac{1}{k^2 r^2}\right) \frac{\sin \theta}{4\pi r} = -\frac{i\eta l dl}{4\pi kr^3} \sin \theta, \\H_\phi &\approx ikl dl \frac{1}{ikr} \frac{\sin \theta}{4\pi r} = \frac{l dl}{4\pi r^2} \sin \theta.\end{aligned}$$

Thus

$$\vec{E} \approx -i \frac{\eta l dl}{4\pi kr^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}), \quad (3.3.7)$$

$$\vec{H} \approx \frac{l dl}{4\pi r^2} \sin \theta \hat{\phi}. \quad (3.3.8)$$

Radiation from a Hertzian Dipole (cont.)

In the far-field, only the $1/r$ terms survive, and the fields simplify to:

$$\vec{E} \approx ik\eta I dl \sin \theta G \hat{\theta}, \quad (3.3.9)$$

$$\vec{H} \approx ik I dl \sin \theta G \hat{\phi}. \quad (3.3.10)$$

In this region, the fields are transverse, mutually orthogonal, and satisfy $\vec{H} = \hat{r} \times \vec{E}/\eta$. In lossless free space k, η are real, while I may be a complex peak-amplitude phasor. Since $\hat{\theta} \times \hat{\phi} = \hat{r}$, the time-averaged Poynting vector of (1.5.6) is

$$\begin{aligned} \vec{S}_{\text{avg}} &= \frac{1}{2} \Re\{\vec{E} \times \vec{H}^*\} \\ &\approx \frac{1}{2} \hat{r} (ik\eta I dl G)(ik I dl G)^* \sin^2 \theta \\ &= \hat{r} \frac{\eta}{2} |k I dl|^2 \sin^2 \theta |G|^2 \\ &= \hat{r} \frac{\eta |k I dl|^2}{32\pi^2 r^2} \sin^2 \theta. \end{aligned} \quad (3.3.11)$$

Radiation from a Hertzian Dipole (cont.)

The angular integral needed for the total radiated power is

$$\int_0^{2\pi} d\phi \int_0^\pi \sin^3 \theta d\theta = 2\pi \left[-\cos \theta + \frac{\cos^3 \theta}{3} \right]_0^\pi = \frac{8\pi}{3}.$$

Therefore

$$\begin{aligned} P_{\text{rad}} &= \int_0^{2\pi} \int_0^\pi \vec{S}_{\text{avg}} \cdot \hat{r} r^2 \sin \theta d\theta d\phi \\ &= \frac{\eta |kl dl|^2}{32\pi^2} \frac{8\pi}{3} = \frac{\eta |kl dl|^2}{12\pi}. \end{aligned} \quad (3.3.12)$$

The radiation intensity is defined as $U(\theta, \phi) = r^2 \vec{S}_{\text{avg}} \cdot \hat{r}$, which for the Hertzian dipole becomes

$$U(\theta) = \frac{\eta |kl dl|^2}{32\pi^2} \sin^2 \theta. \quad (3.3.13)$$

The normalized directive-gain pattern is the ratio of the radiation intensity in a given direction to its spherical average:

$$D(\theta) = \frac{U(\theta)}{P_{\text{rad}}/(4\pi)} = \frac{3}{2} \sin^2 \theta. \quad (3.3.14)$$

Radiation from a Hertzian Dipole (cont.)

Here, the term $P_{\text{rad}}/4\pi$ represents the *average radiated power per unit solid angle*. A solid angle $d\Omega = \sin\theta d\theta d\phi$ measures the angular extent of a surface as seen from the origin, and a full sphere subtends a total solid angle of 4π steradians. This shows that the Hertzian dipole radiates with maximum intensity in the plane perpendicular to its axis ($\theta = \pi/2$) and no radiation along the axis itself ($\theta = 0$ or π). Its directivity, the maximum of $D(\theta)$, is $D_0 = 3/2$. The three-dimensional directivity pattern of a Hertzian dipole is illustrated in Fig. 3.1.

From the duality principle, the far field of a magnetic dipole source can be expressed as

$$\vec{E} \approx -ikI_m dl \sin\theta G\hat{\phi}, \quad (3.3.15)$$

$$\vec{H} \approx ik\frac{I_m}{\eta} dl \sin\theta G\hat{\theta}, \quad (3.3.16)$$

where $I_m dl$ is the magnetic-current moment (units V m). No fundamental magnetic-current source has been observed, but equivalent magnetic currents are useful mathematical sources. An electrically small electric-current loop of area S , current I_ℓ , and magnetic dipole moment $\vec{m} = I_\ell S\hat{z}$ has the same leading far field when

$$I_m dl = i\omega\mu I_\ell S = ik\eta I_\ell S.$$

Radiation from a Hertzian Dipole (cont.)

Indeed, substitution in the preceding magnetic-current fields gives

$$\vec{E} \approx k^2 \eta I_\ell S G \sin \theta \hat{\phi}, \quad \vec{H} \approx -k^2 I_\ell S G \sin \theta \hat{\theta},$$

the small-loop far field. The equivalence here is a dipole-order far-field equivalence, not the assertion that a finite loop is an exact magnetic-current element at every distance.

Radiation from a Hertzian Dipole

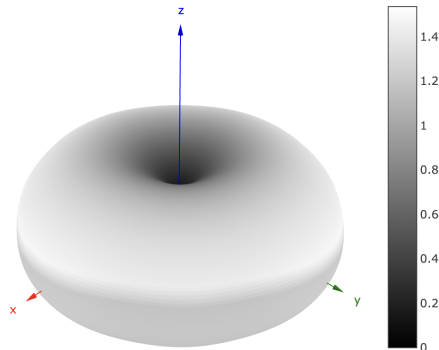


Figure 3.1: Hertzian-dipole directivity $D(\theta) = \frac{3}{2} \sin^2 \theta$: the z-axis is a null and the equatorial maximum is $D_0 = 1.5$.

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Image theory constructs a source configuration whose field satisfies the same differential equation, conductor boundary condition, and condition at infinity as the original problem; uniqueness then identifies the two fields in the region of interest. As a first example, place a charge q at $(0, 0, h)$, $h > 0$, above a grounded PEC plane $z = 0$. Put an image charge $-q$ at $(0, 0, -h)$. In $z > 0$ the proposed potential is

$$\varphi(x, y, z) = \frac{q}{4\pi\epsilon} \left[\frac{1}{\sqrt{x^2 + y^2 + (z - h)^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z + h)^2}} \right]. \quad (3.4.1)$$

At the conductor,

$$\varphi(x, y, 0) = \frac{q}{4\pi\epsilon} \left[\frac{1}{\sqrt{x^2 + y^2 + h^2}} - \frac{1}{\sqrt{x^2 + y^2 + h^2}} \right] = 0.$$

Image Theory (cont.)

The image singularity lies outside the physical half-space. Consequently, in $z > 0$

$$\nabla^2 \varphi = -\frac{q}{\epsilon} \delta(x) \delta(y) \delta(z - h),$$

and $\nabla^2 \varphi = 0$ only away from the real charge. Thus the proposed potential satisfies Poisson's equation at the source, Laplace's equation elsewhere, the grounded-PEC boundary value, and decay at infinity. These are the hypotheses that make the upper-half-space solution unique.

For time-harmonic sources the same method applies when the conductor is ideal and the image orientation is chosen to enforce the boundary condition at every point on the plane. A PEC requires $\hat{n} \times \vec{E} = 0$; a PMC requires $\hat{n} \times \vec{H} = 0$.

The image rules must be checked with the exact dipole field, not merely with the far-field approximation (3.3.9). For a current moment $I d\vec{l}$, rotationally generalizing the exact components derived above gives

$$\vec{E}(\hat{p}, \vec{R}) = \alpha(R) \hat{p} + \beta(R) (\hat{p} \cdot \hat{R}) \hat{R}, \quad (3.4.2)$$

Image Theory (cont.)

where $G_R = e^{-ikR}/(4\pi R)$ and

$$\begin{aligned}\alpha(R) &= \eta l \, dl \, G_R \left(-ik - \frac{1}{R} + \frac{i}{kR^2} \right), \\ \beta(R) &= \eta l \, dl \, G_R \left(ik + \frac{3}{R} - \frac{3i}{kR^2} \right).\end{aligned}\tag{3.4.3}$$

To see these coefficients directly, write $\hat{p} = \cos \theta \, \hat{R} - \sin \theta \, \hat{\theta}$. Equation (3.4.2) then has components

$$E_R = (\alpha + \beta) \cos \theta, \quad E_\theta = -\alpha \sin \theta.$$

Matching them to the exact E_r, E_θ above gives

$$\begin{aligned}-\alpha &= ik\eta l \, dl \, G_R \left(1 + \frac{1}{ikR} - \frac{1}{k^2 R^2} \right), \\ \alpha + \beta &= \frac{2\eta l \, dl}{R} G_R \left(1 + \frac{1}{ikR} \right),\end{aligned}$$

whose solution is precisely (3.4.3).

Image Theory (cont.)

Let the real source be at $(0, 0, h)$, its image at $(0, 0, -h)$, and the field point be $(x, y, 0)$. Define the common in-plane vector

$$\vec{t} = \frac{x\hat{x} + y\hat{y}}{R}, \quad R = \sqrt{x^2 + y^2 + h^2}.$$

The two source-to-field unit vectors and Green functions obey

$$\hat{R}_+ = \vec{t} - \frac{h}{R}\hat{z}, \quad \hat{R}_- = \vec{t} + \frac{h}{R}\hat{z}, \quad G_{R_+} = G_{R_-};$$

therefore the same scalars $\alpha(R)$ and $\beta(R)$ apply to both fields.

For a vertical electric dipole, choose $\hat{p}_+ = \hat{p}_- = \hat{z}$. Taking the in-plane part of the sum in (3.4.2),

$$(\vec{E}_+ + \vec{E}_-)_{\parallel} = \alpha(\vec{0} + \vec{0}) + \beta \left[\left(-\frac{h}{R}\right) \vec{t} + \left(\frac{h}{R}\right) \vec{t} \right] = \vec{0}.$$

Thus a vertical electric dipole has a same-oriented image above a PEC. For any horizontal orientation $\hat{p} \cdot \hat{z} = 0$, choose $\hat{p}_+ = \hat{p}$ and $\hat{p}_- = -\hat{p}$. Since

$$\hat{p}_+ \cdot \hat{R}_+ = \hat{p} \cdot \vec{t}, \quad \hat{p}_- \cdot \hat{R}_- = -\hat{p} \cdot \vec{t},$$

Image Theory (cont.)

the tangential sum is

$$(\vec{E}_+ + \vec{E}_-)_{\parallel} = \alpha(\hat{p} - \hat{p}) + \beta [(\hat{p} \cdot \vec{t})\vec{t} - (\hat{p} \cdot \vec{t})\vec{t}] = \vec{0}.$$

Hence a horizontal electric dipole has an oppositely oriented image. These cancellations use all $1/R$, $1/R^2$, and $1/R^3$ terms and hold at every nonsource point of the plane. The exact PEC boundary condition and the outgoing radiation condition now justify the uniqueness argument in the upper half-space.

Duality supplies all remaining cases. In the following table *same* and *opposite* refer to the orientation of the image moment relative to the source moment:

	$\perp$	$\parallel$
PEC, electric	+	-
PEC, magnetic	-	+
PMC, electric	-	+
PMC, magnetic	+	-

Here $+$ means the same image orientation and $-$ the opposite orientation; $\perp$ and $\parallel$ refer to the plane. Figure 3.2 summarizes these four sign rules.

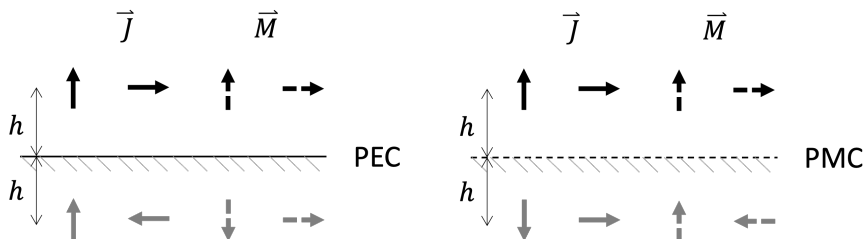


Figure 3.2: Image orientations for electric-current moments $\vec{J}$ and magnetic-current moments $\vec{M}$ above PEC and PMC planes. Black arrows are the real moments, gray arrows are their images, and dashed arrows denote magnetic moments; the orientations agree with the sign table in the text.

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Radiation from an Aperture

When an electromagnetic wave passes through an opening in a conducting screen, the field in the observation half-space can be represented by equivalent currents on the aperture.

Let the aperture lie in $z = 0$, let the desired radiated field occupy $z > 0$, and choose $\hat{n} = +\hat{z}$. With this orientation, Love's equivalent currents are

$$\vec{J}_s = \hat{z} \times \vec{H}_a, \quad \vec{M}_s = -\hat{z} \times \vec{E}_a, \quad (3.5.1)$$

where $\vec{E}_a, \vec{H}_a$ are the tangential aperture fields. Reversing the chosen normal (and the retained half-space) reverses both current signs. On the opaque PEC, $\vec{E}_t = 0$, so $\vec{M}_s = 0$, whereas $\vec{J}_s = \hat{z} \times \vec{H}$ need not vanish until the PEC-replacement step below. To simplify the analysis, consider the transformations in Fig. 3.3. From the surface equivalence principle (1.5.33a), the PEC in the original problem is first replaced by surface currents on the z -plane. Since the tangential electric field on the PEC is zero, the corresponding equivalent magnetic surface current is also zero. In the second step, the surface equivalence principle (1.5.33b) is applied to replace V_1 with a PEC, and the equivalent electric surface current is set to zero in the original PEC region. At this

Radiation from an Aperture (cont.)

stage, only the magnetic surface current remains on the aperture. Finally, applying the image theory removes the PEC and doubles the equivalent magnetic surface current:

$$\vec{M}_s = -2\hat{z} \times \vec{E}_a, \quad (3.5.2)$$

yielding the final representation of the problem. The doubling follows directly from the image configurations summarized in Fig. 3.2: since $\vec{M}_s = -\hat{z} \times \vec{E}_a$ is tangential to the $z = 0$ plane, it is a *horizontal* magnetic current, whose image in a PEC has the *same* orientation. The source and its image therefore reinforce rather than cancel, and removing the conductor leaves a single magnetic current of twice the strength radiating into free space. Substituting $\vec{M}_s = -2\hat{z} \times \vec{E}_a$ into the magnetic-current term of (3.2.13), with $-\hat{r} \times \vec{M}_s = 2\hat{r} \times (\hat{z} \times \vec{E}_a)$, gives

$$\vec{E} \approx -ik \frac{e^{-ikr}}{4\pi r} \int_{S_A} 2\hat{r} \times (\hat{z} \times \vec{E}_a) e^{ik\hat{r} \cdot \vec{r}'} ds',$$

so the factor of 2 is absorbed into the prefactor, replacing $1/(4\pi r)$ by $1/(2\pi r)$ in (3.5.3).

Radiation from an Aperture

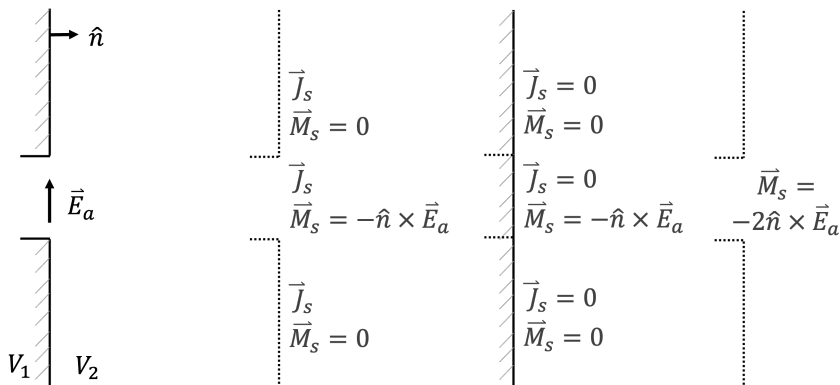


Figure 3.3: Aperture equivalence sequence for the retained half-space: replace the screen by equivalent currents, restore a PEC so only $\vec{M}_s = -\hat{n} \times \vec{E}_a$ remains on the aperture, and then remove the PEC by doubling the same-oriented magnetic current.

Radiation from an Aperture

The far-field electric field follows from (3.2.13) applied to this surface magnetic current:

$$\vec{E} \approx -ik \frac{e^{-ikr}}{2\pi r} \int_{S_A} \hat{r} \times (\hat{z} \times \vec{E}_a) e^{ik\hat{r} \cdot \vec{r}'} ds', \quad (3.5.3)$$

where S_A denotes the aperture area.

Consider a rectangular aperture of dimensions a and b along the x - and y -axes. As an analytically tractable aperture-field model, prescribe $\vec{E}_a = E_0 \hat{y}$ uniformly over the opening and zero outside. A normally incident uniform plane wave motivates this model, but illumination alone does not make the exact aperture field uniform: edge diffraction and coupling generally modify it. Under the stated uniform-field approximation, (3.5.3) can be evaluated explicitly. To reduce the integrand, apply the BAC–CAB identity $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$ with $\vec{a} = \hat{r}$, $\vec{b} = \hat{z}$, and $\vec{c} = \vec{E}_a = E_0 \hat{y}$:

$$\hat{r} \times (\hat{z} \times \vec{E}_a) = E_0 [\hat{z}(\hat{r} \cdot \hat{y}) - \hat{y}(\hat{r} \cdot \hat{z})].$$

Resolving the Cartesian unit vectors into the spherical basis,

$$\hat{y} = \sin \theta \sin \phi \hat{r} + \cos \theta \sin \phi \hat{\theta} + \cos \phi \hat{\phi},$$

$$\hat{z} = \cos \theta \hat{r} - \sin \theta \hat{\theta},$$

Radiation from an Aperture (cont.)

so that the two projections are $\hat{r} \cdot \hat{y} = \sin \theta \sin \phi$ and $\hat{r} \cdot \hat{z} = \cos \theta$. Substituting these,

$$\begin{aligned}\hat{r} \times (\hat{z} \times \vec{E}_a) = E_0 & \left[\sin \theta \sin \phi (\cos \theta \hat{r} - \sin \theta \hat{\theta}) \right. \\ & - \cos \theta (\sin \theta \sin \phi \hat{r} + \cos \theta \sin \phi \hat{\theta} \\ & \left. + \cos \phi \hat{\phi}) \right].\end{aligned}$$

The two $\hat{r}$ contributions, $\sin \theta \cos \theta \sin \phi$ and $-\cos \theta \sin \theta \sin \phi$, cancel, confirming that the far field is transverse. Collecting the $\hat{\theta}$ terms, $-\sin^2 \theta \sin \phi - \cos^2 \theta \sin \phi = -\sin \phi$, and the $\hat{\phi}$ term, $-\cos \theta \cos \phi$, leaves

$$\hat{r} \times (\hat{z} \times \vec{E}_a) = -E_0 (\sin \phi \hat{\theta} + \cos \theta \cos \phi \hat{\phi}).$$

Define the two direction cosines $u = \sin \theta \cos \phi$ and $v = \sin \theta \sin \phi$. The leading factor $-ik$ in (3.5.3) flips this sign, and the constant polarization vector factors out of the surface integral, giving:

$$\begin{aligned}\vec{E} \approx ikE_0 \frac{e^{-ikr}}{2\pi r} \int_{-a/2}^{a/2} \int_{-b/2}^{b/2} & \begin{pmatrix} \sin \phi \hat{\theta} \\ + \cos \theta \cos \phi \hat{\phi} \end{pmatrix} \\ & e^{ik(x'u+y'v)} dx' dy',\end{aligned}\tag{3.5.4}$$

Radiation from an Aperture (cont.)

Since the polarization vector $\sin \phi \hat{\theta} + \cos \theta \cos \phi \hat{\phi}$ does not depend on the integration variables x' and y' , it factors out of the integral, which separates into a product of two one-dimensional integrals. Each is elementary:

$$\begin{aligned}\int_{-a/2}^{a/2} e^{ikux'} dx' &= \left. \frac{e^{ikux'}}{iku} \right|_{-a/2}^{a/2} = \frac{e^{ikua/2} - e^{-ikua/2}}{iku} \\ &= \frac{2 \sin\left(\frac{kua}{2}\right)}{ku} = a \operatorname{sinc}\left(\frac{ka}{2} u\right),\end{aligned}$$

and, in the same way, $\int_{-b/2}^{b/2} e^{ikvy'} dy' = b \operatorname{sinc}(kbv/2)$. Here $\operatorname{sinc} x = \sin x/x$, with its continuous limiting value $\operatorname{sinc} 0 = 1$. Multiplying the two results gives

$$\begin{aligned}\vec{E} &\approx ikE_0 ab \frac{e^{-ikr}}{2\pi r} \begin{pmatrix} \sin \phi \hat{\theta} \\ + \cos \theta \cos \phi \hat{\phi} \end{pmatrix} \\ &\quad \operatorname{sinc}\left(\frac{ka}{2} u\right) \operatorname{sinc}\left(\frac{kb}{2} v\right),\end{aligned}\tag{3.5.5}$$

Radiation from an Aperture (cont.)

The two sinc factors are the rectangular-aperture transform, but they are not the entire vector pattern. The polarization (obliquity) factor has squared magnitude

$$|\sin \phi \hat{\theta} + \cos \theta \cos \phi \hat{\phi}|^2 = \sin^2 \phi + \cos^2 \theta \cos^2 \phi = 1 - u^2.$$

Thus, apart from angle-independent constants, the complete power pattern in the retained observation half-space $0 \leq \theta \leq \pi/2$ is

$$(1 - u^2) \operatorname{sinc}^2\left(\frac{ka}{2}u\right) \operatorname{sinc}^2\left(\frac{kb}{2}v\right).$$

Further Reading

The exposition of radiation solutions within this chapter primarily draws upon Sheng's works [24, 25]. For comprehensive details regarding notations employing dyadic Green functions, the reader is directed to pertinent chapters in Pathak's text [23] and Tai's authoritative book on this topic [28]. The Stratton–Chu formulation is extensively presented in Ishimaru's treatise [12]. The application of the equivalence principle to aperture radiation is also demonstrated in S. K. Jeng's paper [14], which shows how the aperture field due to a plane wave incident on a cavity-backed slit can be obtained, and how the corresponding radiation from the aperture, modeled as a magnetic current, may be computed using the same method described in this section.

Certain advanced topics, including Liénard–Wiechert potentials and radiation from moving charges, fall outside the scope of this discussion. For these subjects, readers are encouraged to consult canonical graduate-level electrodynamics textbooks such as that by Zangwill [31], or specialized volumes on radiation, such as Smith [26].

- 1 Verify (3.1.28) and (3.1.29) by applying the duality transform to (3.1.26) and (3.1.27).
- 2 Following Subsection 3.1.1, first consider magnetic sources with electric sources absent. The electric flux density may be written using the electric vector potential

$$\vec{D} = -\vec{\nabla} \times \vec{F}.$$

Define the magnetic field in terms of the electric vector potential and magnetic scalar potential by

$$\vec{H} = -\vec{\nabla} \vartheta - i\omega \vec{F}.$$

Assuming that the Lorenz condition holds for electric vector potentials,

$$\vec{\nabla} \cdot \vec{F} = -i\omega\mu\epsilon\vartheta.$$

Then use superposition of the electric-source potentials $(\vec{A}, \varphi)$ and magnetic-source potentials $(\vec{F}, \vartheta)$ to show that, when both source types are present, the fields have the general forms

$$\vec{E} = -i\omega\vec{A} + \frac{1}{i\omega\mu\epsilon}\vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \frac{1}{\epsilon}\vec{\nabla} \times \vec{F},$$

Problems (cont.)

$$\vec{H} = -i\omega\vec{F} + \frac{1}{i\omega\mu\epsilon}\vec{\nabla}(\vec{\nabla}\cdot\vec{F}) + \frac{1}{\mu}\vec{\nabla}\times\vec{A}.$$

- 3 Show that (3.1.32) is equal to

$$\begin{aligned} & -ik\eta \int_{V'} \left[\left(\vec{I} + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \right) G(\vec{r}; \vec{r}') \right] \cdot \vec{J}(\vec{r}') dv' \\ & = -ik\eta \int_{V'} \bar{\bar{G}}(\vec{r}; \vec{r}') \cdot \vec{J}(\vec{r}') dv', \end{aligned}$$

where

$$\bar{\bar{G}}(\vec{r}; \vec{r}') = \left[\left(\vec{I} + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \right) G(\vec{r}; \vec{r}') \right]$$

is the conventional electric *dyadic Green function*.

- 4 Complete the intermediate steps in (3.3.3)–(3.3.6).

References



George B. Arfken, Hans J. Weber, and Frank E. Harris.
Mathematical Methods for Physicists: A Comprehensive Guide.
Academic Press, 7th edition, 2011.
Electronic edition.



Constantine A. Balanis.
Advanced Engineering Electromagnetics.
Wiley, 2nd edition, 2012.



Kasra Barkeshli.
Advanced Electromagnetics and Scattering Theory.
Springer, 1st edition, 2015.
Edited by Sina Khorasani.



Craig F. Bohren and Donald R. Huffman.
Absorption and Scattering of Light by Small Particles.
Wiley Science Series. John Wiley & Sons, 2008.
Revised electronic reprint of the 1983 original.

References (cont.)



Weng Cho Chew.

Waves and Fields in Inhomogeneous Media.

IEEE Press Series on Electromagnetic Wave Theory. Wiley-IEEE Press, 1st edition, 1999.

IEEE Press reprint of the 1990 version with corrections and minor updating.



Phillip C. Clemmow.

The Plane Wave Spectrum Representation of Electromagnetic Fields, volume 12 of *International Series of Monographs in Electromagnetic Waves*.

Elsevier Science, 2013.

Electronic reissue of the 1966 Pergamon Press first edition.



Robert E. Collin.

Field Theory of Guided Waves.

IEEE Press, 2nd edition, 1990.

References (cont.)



Georges A. Deschamps.

Electromagnetics and differential forms.

Proceedings of the IEEE, 69(6):676–696, June 1981.



NIST Digital Library of Mathematical Functions.

<https://dlmf.nist.gov/>, Release 1.2.7 of 2026-06-15.

F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.



Stuart Gregson, John McCormick, and Clive Parini.

Principles of Planar Near-Field Antenna Measurements.

Number 53 in IET Electromagnetic Waves Series. Institution of Engineering and Technology, 2007.



Roger F. Harrington.

Time-Harmonic Electromagnetic Fields.

IEEE Press Series on Electromagnetic Wave Theory. IEEE Press, 2001.

Classic reissue of the 1961 original.

References (cont.)



Akira Ishimaru.

Electromagnetic Wave Propagation, Radiation, and Scattering: From Fundamentals to Applications.

Wiley-IEEE Press, 2nd edition, 2017.



John David Jackson.

Classical Electrodynamics.

Wiley India, 3rd edition, 2012.

Indian reprint of the third edition.



Shyh-Kang Jeng.

Scattering from a cavity-backed slit in a ground plane—TE case.

IEEE Transactions on Antennas and Propagation, 38(10):1523–1529, October 1990.



Jian-Ming Jin.

Theory and Computation of Electromagnetic Fields.

Wiley-IEEE Press, 2nd edition, 2015.

References (cont.)



Milton Kerker.

The Scattering of Light and Other Electromagnetic Radiation, volume 16 of *Physical Chemistry: A Series of Monographs*.

Academic Press, 2013.

Series edited by Ernest M. Loebl; electronic reissue of the 1969 first edition.



Jin Au Kong.

Theory of Electromagnetic Waves.

Wiley-Interscience, New York, 1975.



Jin Au Kong.

Electromagnetic Wave Theory.

EMW Publishing, Cambridge, MA, 2008.



Ismo V. Lindell.

Differential Forms in Electromagnetics.

Wiley-IEEE Press, 2004.

References (cont.)



Kyle A. Novak and Laura J. Fox.

Special Functions of Mathematical Physics: A Tourist's Guidebook.

Equal Share Press, LLC, 2019.



Lukas Novotny and Bert Hecht.

Principles of Nano-Optics.

Cambridge University Press, 2nd edition, 2012.



Andrey V. Osipov and Sergei A. Tretyakov.

Modern Electromagnetic Scattering Theory with Applications.

John Wiley & Sons, 1st edition, 2017.



Prabhakar H. Pathak and Robert J. Burkholder.

Electromagnetic Radiation, Scattering, and Diffraction.

Wiley-IEEE Press, 2021.

References (cont.)



Xin-Qing Sheng.

Electromagnetic Theory, Calculations, and Applications.

Tsinghua University Press, Beijing, 2nd edition, 2023.

In Chinese.



Xin-Qing Sheng and Wei Song.

Essentials of Computational Electromagnetics.

Wiley, 2012.



Glenn S. Smith.

An Introduction to Classical Electromagnetic Radiation.

Cambridge University Press, 1st edition, 1997.



Carlo G. Someda.

Electromagnetic Waves.

CRC Press, 2nd edition, 2006.

References (cont.)



Chen-To Tai.

General Vector and Dyadic Analysis: Applied Mathematics in Field Theory.
Wiley-IEEE Press, 2nd edition, 1997.



H. C. van de Hulst.

Light Scattering by Small Particles.
Dover Publications, 1981.
Reprint of the 1957 Wiley edition.



Pochi Yeh.

Optical Waves in Layered Media.
Wiley Series in Pure and Applied Optics. Wiley, 2005.
Paperback reissue of the 1988 original.



Andrew Zangwill.

Modern Electrodynamics.
Cambridge University Press, 2013.